

ENVS 193DS Final

[Github repository](#)

```
# reading in packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(here)
```

here() starts at /Users/luciamurillo/github/ENVS-193DS_spring-2026_final

```
library(ggeffects)
library(MuMIn)

# new object `nests` from csv in `data` folder
# using "here" to identify location
nests <- read_csv(here("data", "nest_data_final.csv"))
```

```
Rows: 299 Columns: 10
-- Column specification -----
Delimiter: ","
chr (3): alive, bird_species, ant_species
dbl (7): height_cm, diameter_mm, case_control, case, canopy_width, canopy_le...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

** make invisible!! **
```

Problem 1: Research writing

(a)

In part 1 they used a linear model.
In part 2 they used a Pearson's test.

(b)

We found that the distance to the water's edge tends to predict the probability that American Avocets would use a habitat. ***

We also found a relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and the type of habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat). ****

(c)

One additional variable that could influence the probability of American Avocet microhabitat use is the type of habitat, which is a categorical variable. They may have preferred food sources or shelter in some habitats and not others, which could influence the probability of microhabitat use. This was a relationship that was already explored ***** maybe not since alr explored??

Problem 2: Data analysis

MAKE SURE TO MAKE INVISIBLE

(a)

```
# creating new object from nests dataframe
nests_clean <- nests |>
  # creating new column with full names of ants
  mutate(species_name = recode_values(
    ant_species,
    "RRB" ~ "C. mimosae",
    "AB" ~ "C. sjostedti",
    "BBR" ~ "C. nigriceps"
  )) |>
  # reordering species levels
  mutate(species_name =
    as_factor(species_name),
    species_name =
      fct_relevel(species_name,
        "C. sjostedti",
        "C. mimosae",
        "C. nigriceps"
      )) |>
  # selecting columns of interest (species name, case_control, and height_cm)
  select(species_name, case_control, height_cm) |>
  # dropping N/A values from species name column
  drop_na(species_name)
```

```
# checking structure of data frame
str(nests_clean)
```

```
tibble [270 x 3] (S3: tbl_df/tbl/data.frame)
 $ species_name: Factor w/ 3 levels "C. sjostedti",...: 2 2 2 2 1 2 3 2 1 3 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
```

**** MAKE SURE STRUCTURE LOOKS GOOD! ****

```
# displaying from nests_clean
slice_sample(
  nests_clean,
  # displaying 10 rows
  n = 10)
```

```
# A tibble: 10 x 3
  species_name case_control height_cm
  <fct>         <dbl>      <dbl>
1 C. mimosae      0        215
2 C. mimosae      0        343
3 C. mimosae      0        446
4 C. mimosae      0        182
5 C. mimosae      0        360.
6 C. mimosae      1        201
7 C. mimosae      0        125
8 C. mimosae      0        212.
9 C. nigriceps    0        167
10 C. mimosae     0        104
```

(b)

In this data set, the 0s and 1s represent whether a tree contained a nest (1) or was an “available” tree for comparison (0). **expand on this?**

(c)

Tree height is a continuous variable. I would expect that as height increases, the probability of nesting would also increase. This is supported by the article, which says that for every 1m increase in height, the odds that birds nested in a tree increased by 20% (Results section, paragraph 2).

Acacia ant species are categorical variables. I would expect that in trees with *C. nigriceps* and *C. mimosae*, there would be a higher probability of nest occurrence. Based on the article,

it seems that these ants are the most aggressive. This seems to increase probability of nests because these ants may help reduce the risk of nest predation (Discussion section of the article in paragraph 1).

(d)

Model number	Tree height	Ant species	Predictor list
0			no predictors (null model)
1	X	X	all predictors (full model)
2	X		Tree height
3		X	Ant species

(e)

```
# model 0: null model
model0 <- lm(
  # formula: no predictors
  case_control ~ 1,
  # from clean data frame
  data = nests_clean
)
# model 1: all predictors
model1 <- lm(
  # formula: all predictors
  case_control ~ species_name + height_cm,
  # from clean data frame
  data = nests_clean
)
# model 2: tree height
model2 <- lm(
  # formula: height only
  case_control ~ height_cm,
  # from clean data frame
  data = nests_clean
)
# model 3: ant species
model3 <- lm(
  # formula: species only
  case_control ~ species_name,
```

```
# from clean data frame
data = nests_clean
)
```

(f)

(g)

(h)

(i)

(j)

(k)

Problem 3: Affective and exploratory visualizations

(a)

The exploratory visualizations I made in Homework 2 are very different from the affective visualization I made for Homework 3. The exploratory visualizations were made to get information across in a simple and understandable way. The affective visualization contains data, but that is not the only goal. It is more about being visually interesting with the explanation of the data being represented being on another sheet of paper.

They are all similar in that they come from the same dataset. The affective visualization is also similar to the boxplot from the exploratory visualization. They both display the productivity at different locations.

In the affective visualization, the mean productivity levels are displayed, which show a similar pattern to the boxplot, which shows the median productivity among other things. The productivity based on the time of start does not share any patterns with the other two that I can determine.

In week 9, my group thought my idea was good. However, some members suggested that I add other variables from my data. I did not end up doing this because I was unsure how to do that in a way that would make sense logically and artistically.

(b)